

2026 XUPT Campus Mathematical Contest in Modeling

Problem X: [Title Here]

Team Number: _____

2026 年 5 月 5 日

摘要

Abstract: Briefly describe the problem background, methodology, key results, and conclusions in one paragraph (approximately 300–500 words).

Keywords: keyword1; keyword2; keyword3

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1 Problem Restatement

1.1 Background

Restate the problem background concisely.

1.2 Problem Analysis

- Problem 1: ...
- Problem 2: ...

2 Assumptions and Notations

2.1 Fundamental Assumptions

1. Assumption 1.
2. Assumption 2.

2.2 Notations

表 1: Symbol Descriptions

Symbol	Description	Unit
x	Decision variable	–
t	Time	s
λ	Lagrange multiplier	–

3 Model Establishment and Solution

3.1 Problem 1: Model Building

A sample equation:

$$\min_{x \in \mathbb{R}^n} f(x) \quad \text{s.t.} \quad g_i(x) \leq 0, \quad i = 1, \dots, m \quad (1)$$

3.2 Problem 1: Algorithm Design

Describe the solver or heuristic used.

3.3 Problem 1: Results

4 Sensitivity Analysis

Analyze how the solution changes with perturbations to key parameters.

5 Model Evaluation

5.1 Strengths

- ...

5.2 Weaknesses

- ...

5.3 Extensions

Potential improvements and generalizations.

6 Conclusion

Summarize findings.

参考文献

[1] Author A. *Title of Paper*. Journal Name, Year.

[2] Author B. *Book Title*. Publisher, Year.

A Source Code

Listing 1: Main solver

```
1 """
2 2026 XUPT Campus MCM - Main solver entry point.
3 """
4 from pathlib import Path
5
6 import numpy as np
7 import pandas as pd
```

```
8 import matplotlib.pyplot as plt
9
10 plt.rcParams["font.sans-serif"] = ["SimHei", "Microsoft YaHei", "DejaVu
    Sans"]
11 plt.rcParams["axes.unicode_minus"] = False
12
13 ROOT = Path(__file__).resolve().parent.parent
14 DATA_DIR = ROOT / "data"
15 FIG_DIR = ROOT / "figures"
16 FIG_DIR.mkdir(exist_ok=True)
17
18
19 def load_data(name: str) -> pd.DataFrame:
20     return pd.read_csv(DATA_DIR / name)
21
22
23 def main() -> None:
24     print("XUPT MCM 2026 - solver skeleton ready")
25
26
27 if __name__ == "__main__":
28     main()
```